

2022 JUNIOR CANADIAN CHEMISTRY OLYMPIAD

The Periodic Table/Data Sheet may be used for all parts of the contest, and non-programmable calculators are allowed.

考试期间允许使用化学元素周期表、数据表以及非编程计算器。

Write your answers on the Scantron Sheet provided using a 2B pencil. 请将答案用 2B 铅笔正确填涂在答题卡上。

Students are expected to attempt all questions for a complete examination in 1.5 hours. 考试时长为 1.5 小时。

- 1) Which of the following chemical equations is properly balanced?
以下化学方程式配平正确的是哪个?
A) $\text{C}_6\text{H}_{12}\text{O}_6 + 12\text{O}_2 \rightarrow 6\text{H}_2\text{O} + 6\text{CO}_2$
B) $\text{C}_6\text{H}_8 + 6\text{O}_2 \rightarrow 5\text{CO}_2 + 4\text{H}_2\text{O}$
C) $\text{C}_{12}\text{H}_{22}\text{O}_{11} + \text{H}_2\text{O} \rightarrow 2\text{C}_6\text{H}_{12}\text{O}_6$
D) $\text{C}_2\text{H}_2 + 2\text{O}_2 \rightarrow 2\text{CO}_2 + \text{H}_2\text{O}$
E) $\text{C}_{12}\text{H}_{22}\text{O}_{12} + \text{H}_2\text{O} \rightarrow 2\text{C}_6\text{H}_{12}\text{O}_6$
- 2) High Fructose Corn Syrup (HFCS) is used as a sweetener in soft drinks. The type of HFCS that is most commonly used in Coca-Cola is 55 percent fructose and 45 percent glucose (HFCS-55). The glucose and fructose concentrations in Coca-Cola are 0.270M and 0.330M respectively. What is the mass of glucose ($\text{C}_6\text{H}_{12}\text{O}_6$) and fructose ($\text{C}_6\text{H}_{12}\text{O}_6$) in a 375 mL can of Coke?
高果糖玉米糖浆 (HFCS) 用作软饮料中的甜味剂。可口可乐中最常用的 HFCS 类型是 55% 的果糖和 45% 的葡萄糖 (HFCS-55)。可口可乐中的葡萄糖和果糖浓度分别为 0.270 M 和 0.330 M。一罐 375 毫升可乐中葡萄糖 ($\text{C}_6\text{H}_{12}\text{O}_6$) 和果糖 ($\text{C}_6\text{H}_{12}\text{O}_6$) 的质量分别是多少?
A) 20.5 g of fructose and 16.8 g of glucose
20.5 克果糖和 16.8 克葡萄糖
B) 22.3 g of fructose and 18.2 g of glucose
22.3 克果糖和 18.2 克葡萄糖
C) 59.5 g of fructose and 48.6 g of glucose
59.5 克果糖和 48.6 克葡萄糖
D) 33.0 g of fructose and 27.0 g of glucose
33.0 克果糖和 27.0 克葡萄糖
E) 27.5 g of fructose and 22.5 g of glucose
27.5 克果糖和 22.5 克葡萄糖
- 3) The density of sodium chloride at 25°C is 2.17 g/cm³. What is the volume of 1 mole of NaCl at 25°C?
温度为 25°C 时, 氯化钠的密度为 2.17 g/cm³。25°C 时 1 摩尔氯化钠 (NaCl) 的体积是多少?
A) 26.9 cm³ B) 18.0 cm³ C) 126.8 cm³
D) 29.6 cm³ E) 58.4 cm³
- 4) The New Afton mine in British Columbia is found to contain 0.68 grams of gold per ton of ore. If 45.7 tons of ore is collected from New Afton mine, how many moles of gold can be extracted from ore?
不列颠哥伦比亚省的新阿夫矿场中发现含金量为 0.68 克/吨的矿石。如果从新阿夫矿场收集 45.7 吨矿石, 可以从矿石中提取多少摩尔的黄金?
A) 0.015 B) 1.00 C) 0.158 D) 6.12 E) 0.232
- 5) Several general types of chemical reactions can occur based on what happens when going from reactants to products. What is the type of the following reaction?
根据从反应物到产物的变化, 会发生几种常见的化学反应。以下属于哪种反应类型?
$$\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$$

A) Synthesis 化合反应
B) Single replacement 置换反应
C) Decomposition 分解反应
D) Neutralization 中和反应
E) Combustion 燃烧反应

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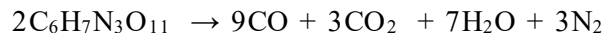
- 6) The cause of anemia, a low level of red blood cells, is often iron deficiency. Pharmaceutical-grade ferric citrate can be prepared from the reaction of a ferric salt with citric acid. What is the iron concentration of a solution prepared with 1.30 g of iron chloride hexahydrate ($\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$) in a 100 mL volumetric flask?

贫血（即红细胞低于正常水平）的原因通常是缺铁。药用级柠檬酸铁可由铁盐与柠檬酸反应制备而成。用 1.30 克六水三氯化铁（ $\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$ ）在 100 mL 容量瓶中制备的溶液中铁的浓度是多少？

- A) $4.81 \times 10^{-1} \text{ M}$ D) $8.01 \times 10^{-2} \text{ M}$
 B) $8.01 \times 10^{-1} \text{ M}$ E) $4.81 \times 10^{-2} \text{ M}$
 C) $4.81 \times 10^{-3} \text{ M}$

- 7) Nitrocellulose (also called guncotton) is a highly flammable compound that can be made by exposing cotton to a mixture of sulfuric and nitric acid. Guncotton can detonate if it is confined, as it would be in the barrel of a gun. The detonation reaction involving nitrocellulose is schematized below:

硝化纤维素（也称硝化棉）是一种高度易燃的化合物，可以通过将棉放在硫酸和硝酸混合物中制成。硝化棉如果被密封，就像放在枪管里一样，可以爆炸。涉及硝化纤维素的爆炸反应方程式如下：



If 100g of guncotton is detonating, how many grams of water is produced?

如果将 100 克硝化棉引爆，会产生多少克的水？

- A) 21.2 B) 6.06 C) 42.4 D) 350 E) 63.1

- 8) A 50.0 mL canister of Freon-12 (CF_2Cl_2) was heated in boiling water (100.0°C) until the canister burst. If the canister was not defective, and had a burst rating of 102.3 atm, what minimum amount of Freon-12 was in the canister? (assume ideal gas behavior)

将置于 50.0 mL 容器内的氟利昂-12（ CF_2Cl_2 ）在沸水

（ 100.0°C ）中加热，直到容器破裂。如果容器没有缺陷，并且爆裂等级为 102.3 atm，那么容器中氟利昂-12 的最小量是多少？（假设为理想气体情况下）

- A) 199 g B) 0.20 g C) 20.2 g D) 19.7 g E) 75.4 g

- 9) Which of the following molecules can form a hydrogen bond?

下列哪种分子可以形成氢键？

- A) NaH B) NH_3 C) CH_3F D) H_2CO E) CH_2Cl_2

- 10) What is the molecular geometry of PCl_5 ?

PCl_5 分子的几何构型是什么？

- A) Trigonal Planar 平面三角形
 B) Trigonal Bipyramidal 三角双锥
 C) Square Pyramidal 四方锥形
 D) Trigonal pyramidal 三棱锥
 E) Tetrahedral 正四面体

- 11) Which of the following element is expected to be the most electronegative?

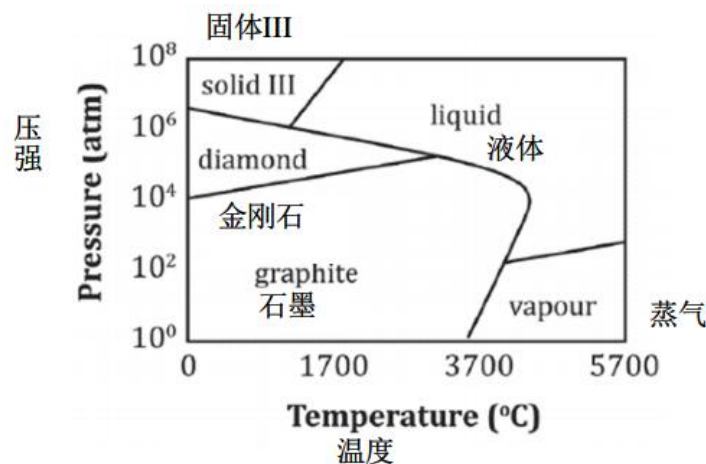
以下哪种元素的电负性最强？

- A) Cl B) C C) Na D) S E) Br

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- 12) Synthetic diamonds have many uses in industry. Diamond is the hardest known naturally occurring material. This property makes diamond the ideal material for machine tools and cutting tools. Synthetic diamonds can be obtained directly from graphite (carbon) under high pressure. The phase diagram of carbon is given below. What phase change occurs by going from 4700 °C to 2700 °C at a pressure of 10 atmosphere?

人造金刚石在工业上有很多用途。金刚石是目前已知最坚硬的天然材料，这使得金刚石成为机床和切削工具的理想材料。人造金刚石可直接将石墨（碳）置于高压条件下获取。碳的相图如下所示。10 个大气压下，温度从 4700°C 到 2700°C 发生的相变是什么？



- | | |
|-----------------|----|
| A) Fusion | 熔解 |
| B) Deposition | 凝华 |
| C) Vaporization | 汽化 |
| D) Freezing | 凝固 |
| E) Sublimation | 升华 |

- 13) Effective nuclear charge (Z_{eff}) is a useful tool for understanding periodicity. Z_{eff} is the net positive charge experienced by an electron in a polyelectronic atom. The term "effective" is used because the positive charge of the nuclei cannot reach a given electron completely, due to the shielding effect of other electrons. Z_{eff} is a theoretically calculated value (not an atomic property) but a good estimate of Z_{eff} can be obtained using Slater's rule. Slater's rule states that $Z_{\text{eff}} = Z - S$ where Z is the atomic number of the atom and S a screening constant. In the second row of the periodic table, the screening value is 0.85 for a core electron and 0.35 for a valence electron. The screening constant (S) is obtained by the summation of all the screening values of other electrons (excluding the electron for which the effective nuclear charge is to be determined). Use Slater's rule to estimate the effective nuclear charge for a valence electron in nitrogen (N).

有效电荷数 (Z_{eff}) 是理解周期性的有用工具。 Z_{eff} 是电子在多电子原子中的净正电荷。使用术语“有效”是因为原子核的正电荷不能完全到达给定的电子，这是由于其他电子的屏蔽效应。 Z_{eff} 是一个理论的计算值（不是原子性质）， Z_{eff} 的估计值可以使用斯莱特法则获取。斯莱特法则规定 $Z_{\text{eff}} = Z - S$ ，其中 Z 是原子的原子序数， S 是屏蔽常数。在周期表的第二行，内层电子的屏蔽常数为 0.85，价电子为 0.35。屏蔽常数 (S) 通过计算其他电子（不包括需要确定有效电荷数的这个电子）的所有屏蔽常数之和得到。使用斯莱特法则估算氮 (N) 中价电子的有效电荷数。

- A) 3.80 B) 5.60 C) 3.55 D) 3.90 E) 5.25

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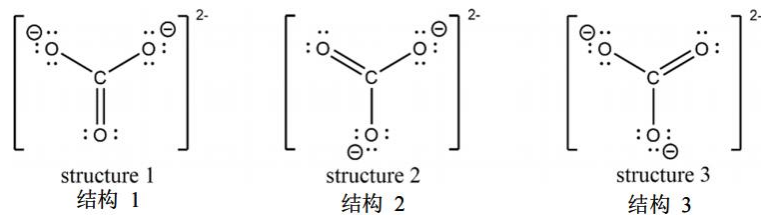
- 14) Calculate ΔH° for the reaction of ice with sodium metal at 0°C (assume the enthalpies given in the table are valid at this temperature):

计算 0°C 下冰与金属钠反应的 ΔH° 。(假设表中给出的焓在此温度下有效):



$\Delta H_f^\circ[\text{Na}^+(\text{aq})]$	-240.1 kJ/mol
$\Delta H_f^\circ[\text{OH}^-(\text{aq})]$	-230.0 kJ/mol
$\Delta H_f^\circ[\text{H}_2\text{O}(\text{l})]$	-285.83 kJ/mol
$\Delta H_{\text{fus}}[\text{H}_2\text{O}(\text{s})]$	6.088 kJ/mol

- A) -185.1 kJ/mol
 B) -178.2 kJ/mol
 C) 185.1 kJ/mol
 D) 178.2 kJ/mol
 E) -749.8 kJ/mol
- 15) Which Lewis structure(s) below is (are) good for carbonate?
 下面哪个/哪些路易斯结构更符合碳酸盐的结构?



- A) Structure 1 only 只有结构 1
 B) Structure 2 only 只有结构 2
 C) Structure 3 only 只有结构 3
 D) All the structures shown 以上所有结构
 E) None of the structures shown 以上都不是

- 16) Which of the following atoms or ions has the smallest radius?

下面哪个原子或离子的半径最小?

- A) Xe B) Fr^+ C) Cs^+ D) Rn E) I^-

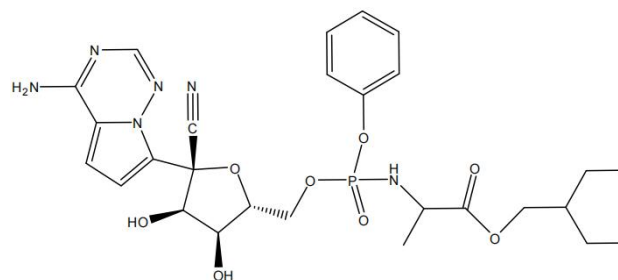
- 17) Which of the following atoms has the largest first ionization energy?

下面哪个原子的第一电离能最大?

- A) Si B) Ca C) K D) Be E) Cl

- 18) The use of antiviral drugs to treat the coronavirus disease (COVID-19) has been controversial. Despite the controversy, the antiviral drug shown below was approved by the FDA for the treatment of COVID-19. The drug was also approved in Canada a few months later. Which of the following molecular formula is the correct one for the antiviral drug shown below?

使用抗病毒药物治疗新型冠状病毒肺炎 (COVID-19) 一直存在争议。尽管如此, 食品药品监督管理局 (FDA) 批准了以下抗病毒药物可用于治疗新型冠状病毒肺炎。数月后, 该药物也在加拿大获得批准。下面哪项是抗病毒药物的正确分子式?

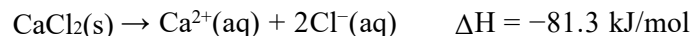


- A) $\text{C}_{27}\text{H}_{34}\text{N}_5\text{O}_8\text{P}$ D) $\text{C}_{27}\text{H}_{35}\text{N}_6\text{O}_8\text{P}$
 B) $\text{C}_{25}\text{H}_{34}\text{N}_6\text{O}_8\text{P}$ E) $\text{C}_{27}\text{H}_{35}\text{N}_6\text{O}_7\text{P}$
 C) $\text{C}_{27}\text{H}_{32}\text{N}_6\text{O}_8\text{P}$

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- 19) 25.0 g of calcium chloride (CaCl_2) is added to 50.0 g of ice at 0°C in an insulated container. Use the data below to calculate the final temperature of the system, in $^\circ\text{C}$.

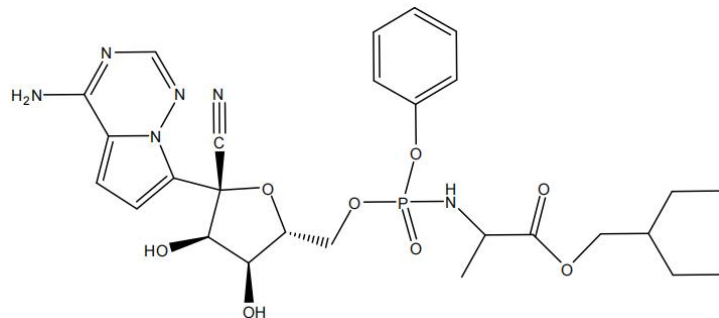
将 25.0 g 氯化钙 (CaCl_2) 加入装有温度为 0°C 的 50.0 g 冰块的绝缘容器内。使用以下数据计算容器内的最终温度。(单位为 $^\circ\text{C}$)



$C_p, \text{mH}_2\text{O}(\text{l})$	$75.38 \text{ JK}^{-1}\text{mol}^{-1}$
$C_p, \text{mH}_2\text{O}(\text{s})$	$37.66 \text{ JK}^{-1}\text{mol}^{-1}$
$\Delta H_{\text{fus}} [\text{H}_2\text{O}(\text{s})]$	6.00 kJ/mol

- A) 15.7 B) 58.8 C) 21.9 D) 7.9 E) 87.5

- 20) The antiviral drug in question 18 has the structure shown as,
第 18 题中的抗病毒药物具有如下所示的结构,



Which of the following correctly lists the organic functional groups presented in the molecule?

以下哪项正确列出了分子中的有机官能团?

- A) Amine, Ester, Ether, Alkene, Nitrile
氨基、酯基、醚、烯烃、腈
- B) Amide, Ketone, Ether, Alkene, Nitrile
酰胺基、酮基、醚、烯烃、腈

- C) Ester, Alcohol, Amine, Nitro, Ether

酯基、醇、氨基、硝基、醚

- D) Ketone, Cyanide, Amide, Benzene, Alcohol

酮基、氰化物、酰胺基、苯、醇

- E) Amine, Ether, Alcohol, Ester, Nitrile

氨基、醚、醇、酯基、腈

- 21) Using your knowledge of organic chemistry, structure of molecule and intermolecular forces to determine which of the following organic compounds has the highest vapor pressure under 1 atm and 20°C .

利用有机化学、分子结构和分子间作用力的知识, 判断下面哪个有机化合物在 1 atm, 20°C 条件下的蒸汽压最高?

- A) Amino-butane 氨基丁烷

- B) Butanol 丁醇

- C) Diethyl ether 二乙醚

- D) Ethanol 乙醇

- E) Ethyl Ethanoate 乙酸乙酯

- 22) Canada is one of the world's largest producers of uranium used in the nuclear power plants. ^{238}U has a half life of $4.468 \times 10^9 \text{ y}$. What would be the rate constant when ^{238}U undergoes radioactive decay? 加拿大是世界上核电站用铀的最大生产国之一。 ^{238}U 的半衰期为 $4.468 \times 10^9 \text{ y}$ 。 ^{238}U 发生放射性衰变时, 速率常数是多少?

- A) $1.55 \times 10^{-10} \text{ y}^{-1}$ B) $1.99 \times 10^{19} \text{ y}^2$ C) $2.24 \times 10^9 \text{ y}$

- D) $2.24 \times 10^{-10} \text{ y}^{-1}$ E) $6.68 \times 10^4 \text{ y}^{1/2}$

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- 23) A solution is made by mixing 8.30 ml of 0.630M NaCl, 4.20 ml of 0.432 M CaCl₂, 5.10 ml of 0.322M of FeCl₃ and 2.40 ml of 0.280M of AgNO₃. The final concentration of chloride ion in the mixture will be the closest to,
- 将 8.30 ml 0.630M 的 NaCl、4.20 ml 0.432M 的 CaCl₂、5.10 ml 0.322M 的 FeCl₃ 和 2.40 ml 0.280M 的 AgNO₃ 混合制成溶液。混合物中氯离子的最终浓度将最接近以下哪个值？

A) 0.312M B) 0.434M C) 0.494M D) 0.656M E) 0.689M

- 24) A company wants to electroplate a spoon with silver metal. What current is needed to electroplate 5.3 g of silver metal from a silver nitrate (AgNO₃) solution if the process runs for 45 minutes?

hint: use the equation $t = \frac{Q}{I}$ where Q is the amount of electricity in coulombs, I is the current in ampere and t is the time in seconds.

一家公司想用银金属电镀勺子。如果整个过程进行 45 分钟，从硝酸银（AgNO₃）溶液中镀出 5.3 克银金属需要多大的电流？

注意：公式 $t = \frac{Q}{I}$ 中， Q 是以库仑为单位的电量， I 是以安培为单位的电流， t 是以秒为单位的时间。

A) 1.755 A B) 0.570 A C) 1.115 A D) 0.897 A E) 105.3 A

- 25) A chemist adds 100 mL of water to 100 mL of a 0.35 M acetic acid solution (pK_a = 4.80). Calculate the pH of the resulting solution.

一位化学家向 100 mL 0.35 M 的乙酸溶液（pK_a = 4.80）中加入 100 mL 水。请计算最终溶液的 pH 值。

A) 2.63 B) 2.78 C) 5.56 D) 5.26 E) 2.98

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Periodic table and data sheet

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1 H 1.008 Hydrogen																	2 He 4.003 Helium				
3 Li 6.941 Lithium	4 Be 9.012 Beryllium															5 B 10.81 Boron	6 C 12.01 Carbon	7 N 14.01 Nitrogen	8 O 16.00 Oxygen	9 F 19.00 Fluorine	10 Ne 20.18 Neon
11 Na 22.99 Sodium	12 Mg 24.31 Magnesium															13 Al 26.98 Aluminium	14 Si 28.09 Silicon	15 P 30.97 Phosphorus	16 S 32.07 Sulfur	17 Cl 35.45 Chlorine	18 Ar 39.95 Argon
19 K 39.10 Potassium	20 Ca 40.08 Calcium	21 Sc 44.96 Scandium	22 Ti 47.87 Titanium	23 V 50.94 Vanadium	24 Cr 52.00 Chromium	25 Mn 54.94 Manganese	26 Fe 55.85 Iron	27 Co 58.93 Cobalt	28 Ni 58.69 Nickel	29 Cu 63.55 Copper	30 Zn 65.38 Zinc	31 Ga 69.72 Gallium	32 Ge 72.61 Germanium	33 As 74.92 Arsenic	34 Se 78.96 Selenium	35 Br 79.90 Bromine	36 Kr 83.80 Krypton				
37 Rb 85.47 Rubidium	38 Sr 87.62 Strontium	39 Y 88.89 yttrium	40 Zr 91.22 Zirconium	41 Nb 92.91 Niobium	42 Mo 95.96 Molybdenum	43 Tc (98) Technetium	44 Ru 101.1 Ruthenium	45 Rh 102.9 Rhodium	46 Pd 106.4 Palladium	47 Ag 107.9 Silver	48 Cd 112.4 Cadmium	49 In 114.8 Indium	50 Sn 118.7 Tin	51 Sb 121.8 Antimony	52 Te 127.6 Tellurium	53 I 126.9 Iodine	54 Xe 131.3 Xenon				
55 Cs 132.9 Cesium	56 Ba 137.3 Barium	57 La 138.9 Lanthanum	72 Hf 178.5 Hafnium	73 Ta 180.9 Tantalum	74 W 183.9 Tungsten	75 Re 186.2 Rhenium	76 Os 190.2 Osmium	77 Ir 192.2 Iridium	78 Pt 195.1 Platinum	79 Au 197.0 Gold	80 Hg 200.6 Mercury	81 Tl 204.4 Thallium	82 Pb 207.2 Lead	83 Bi 209.0 Bismuth	84 Po (209) Polonium	85 At (210) Astatine	86 Rn (222) Radon				
87 Fr (223) Francium	88 Ra (226) Radium	89 Ac (227) Actinium	104 Rf (261) Rutherfordium	105 Db (262) Dubnium	106 Sg (266) Seaborgium	107 Bh (264) Bohrium	108 Hs (277) Hassium	109 Mt (268) Meitnerium	110 Ds (269) Darmstadtium	111 Rg (272) Roentgenium	112 Cn (285) Copernicium	113 Nh (284) Nihonium	114 Fl (289) Flerovium	115 Mc (288) Moscovium	116 Lv (292) Livermorium	117 Ts (294) Tennessine	118 Og (294) Oganesson				

58 Ce 140.1 Cerium	59 Pr 140.9 Praseodymium	60 Nd 144.2 Neodymium	61 Pm (145) Promethium	62 Sm 150.4 Samarium	63 Eu 152.0 Europium	64 Gd 157.3 Gadolinium	65 Tb 158.9 Terbium	66 Dy 162.5 Dysprosium	67 Ho 164.9 Holmium	68 Er 167.3 Erbium	69 Tm 168.9 Thulium	70 Yb 173.0 Ytterbium	71 Lu 175.0 Lutetium
90 Th 232.0 Thorium	91 Pa 231.0 Protactinium	92 U 238.0 Uranium	93 Np 237.0 Neptunium	94 Pu (244) Plutonium	95 Am (243) Americium	96 Cm (247) Curium	97 Bk (247) Berkelium	98 Cf (251) Californium	99 Es (252) Einsteinium	100 Fm (257) Fermium	101 Md (258) Mendelevium	102 No (259) Nobelium	103 Lr (262) Lawrencium

原子质量单位	Atomic mass unit	1.66054×10 ⁻²⁷ kg
阿伏伽德罗常数	Avogadro's number	6.022×10 ²³
电子电荷量	Charge of an electron	1.60218×10 ⁻¹⁹ C
法拉第常数	Faraday's constant	96485 C mol ⁻¹
理想气体常数	Gas constant	8.3145 J K ⁻¹ mol ⁻¹
电子质量	Mass of an electron	9.1094×10 ⁻³¹ kg
光速	Speed of light	2.99792458×10 ⁸ ms ⁻¹
普朗克常数	Plank Constant	6.62637×10 ⁻³⁴ Js

1 atm = 101325 Pa
1 bar = 100000 Pa
0 °C = 273.15 K